

→ MDP on belief space

POMDPs: S - State space $TCS'|S, a$ - Transition Probabilities distribution A - action space $R(S, a)$ - Reward

O - Observations space $ZCO(a, s')$ - Observation probabilities distribution.

Bayesian Belief Update: $b'(s') \propto ZCO(a, s') \sum_s TCS'|S, a b(s)$ $\pi = \pi(CO)$
 $b(s) = P(S|h_t)$ $h_t = (b_0, a_0, o_1, a_1, o_2, \dots)$

POMDP Objective

$$\pi^* = \arg \max_{\pi: B \rightarrow A} E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

Exact POMDP soln: α -Vectors

How do we calculate the optimal action in a POMDP

One-Step utility (reward): $R(b, a) = \vec{R}^a \cdot \vec{b}$

→ One Step α -Vector.

$$\vec{R}^a = \begin{bmatrix} R(s^1, a) \\ R(s^2, a) \\ \vdots \\ R(s^n, a) \end{bmatrix} \quad \vec{b} = \begin{bmatrix} b(s^1) \\ b(s^2) \\ \vdots \\ b(s^n) \end{bmatrix}$$

$\pi_{greedy}(b) = \arg \max_a \vec{R}^a \cdot \vec{b} \rightarrow$ Does not take into account information gained

Conditional Plans: fixed depth history based policies 2 Step: $\pi(C)$

For 1 step: $u^\pi(s) = R(s, \pi(s))$

2 Step: $u^\pi(s) = R(s, \pi(s)) + \gamma \sum_{C'} \left(TCS'|S, \pi(s) \sum_{s'} ZCO(\pi(s), s') u^\pi(s') \right)$

POMDP Value fns:

$$V^*(b) = \max_{a \in A} \alpha^T b$$

→ Point-backup (Γ, b)
 for $a \in A$, belief update
 for $o \in O$, belief update
 $b' \leftarrow \gamma(b, a, o)$
 $Q_{a,b} \leftarrow \arg \max_{s'} R(s, a) + \gamma V^*(b')$
 for $s \in S$
 $Q_{a,b}[s] = R(s, a) + \gamma \dots$
 return $Q_{a,b}$

Offline POMDP Algorithms: Point Based Value Iteration (PBVI) Maintain belief set B , α -Vector set Γ , Point back $\forall b \in B$.

SARSOP: Start w/ $B = \{b_0\}$, $\Gamma =$ blind lower bound, Maintain B of belief tree, Maintain Value bounds $(\Gamma, \text{lower}, \text{upper})$
 • At each node of tree, sample belief points to add to tree, choose a w/ upper bound, choose o w/ lower bound.
 • Prune Tree & α -Vectors to not be dominant.

Computational Complexity: Finite horizon POMDPs, PSPACE complete! Hardest problems that can be solved exactly = Polynomial amount of space. Any algo that solve a general POMDP will have exp. complexity. (linear, quadratic, quadratic)

Formal Approximations (Solve slightly different problem) Certainty Equivalent: $\pi_{CE}(b) = \pi(s(\vec{s}(b)))$, optimal for LQG

QMDP: $T_{QMDP} = \arg \max_{a \in A} \sum_{s \in S} [Q_{MDP}(s, a)]$ Q for certainty Given $(S, A, O, T, Z, R, \gamma)$ (Markov filter)
 MDP. Certainty MDP $\rightarrow (S, A, T, R, \gamma) \rightarrow$ solve $Q^* \approx Q_{MDP}$

→ Bad at costly information gathering & long lasting uncertainty.

Certainty equivalent can refer to other choices of π_s , such as optimal MDP solutions.

→ very optimistic soln
Hindsight Optimization: K scenarios to plan over, pre sample noise vectors w_t^k for reverses fire step in our horizon in every scenario, calculate hindsight $Q, Q_{HS}(s, a) = R(s, a) + \gamma \frac{1}{K} \sum_{k=1}^K \max_{a'} \sum_{s'} \gamma^t R(s_t^k, a_t^k)$

$S_{t+1} = G(S_t^k, a_t^k, w_t^k) \leftarrow$ subject to $\pi_{HS}(b) = \arg \max_{a \in A} \sum_{s \in S} [Q_{HS}(s, a)]$, by pre selecting noise & optimistic action in scenario, we take advantage of pre sample noise.
 $b' = \gamma(b, a, o)$

K-Markov: Solve an MDP where the state is the previous K observations $S_t^k [O_t, O_{t-1}, \dots, O_{t-K+1}]$

Open loop: Open loop feedback control: $T_{OLFC}(b) = \arg \max_{a \in A} \left(\sum_{s \in S} \sum_{t=0}^T \gamma^t R(s_t, a_t) \right)$ can loop



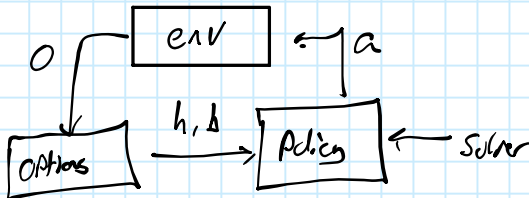
OLFC: Choose sequence of actions that optimizes obj. $E[\cdot]$
 → Good if starting is low, CPB can hold to reduce approximation

Formal Approximate Task		
Certainty Equivalent	Control as if mean is the true state	→ optimal for LQG, very optimistic
QMDP	Full observability after 1 step (hindsight knowledge of state uncertainty)	→ lower bound on true value → optimistic
Hindsight Optimization	Hindsight knowledge of state & action uncertainty	→ lower bound on true value → more optimistic than QMDP
K-Markov	Pretend last K observations make up state & solve MDP	→ Good for deterministic, persistent

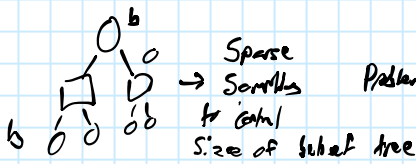
Particle Filter POMDPs: When only estimating the reward, N_p does not need to scale exponentially w/ dimension. OCA's complexity.

Online POMDP Methods:

Sense - Plan - Act Loop:



How should we adapt MCTS for POMDPs?
- MDP on belief space?

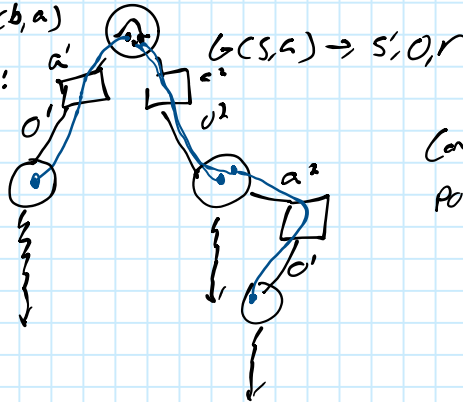


Problem: Belief updates are expensive, number of compute: $O(|S|^2)$

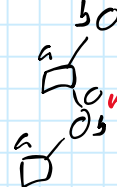
- 1) History
- 2) Belief update: $b_t = P(S_t | h_t)$ b sample state

MCTS on Histories:

Don't keep track of states we switch.



History based Planning!



Continuous POMCP

Observed States POMCROW

Recall: UCB requires each action in a node at least once.
MCTS Q update (backprop)
 $Q(h, a) \leftarrow Q(h, a) + \frac{a - Q(h, a)}{N(h, a)}$
 $Q \leftarrow \max_r \text{Reduct}(S', h, a, o, b-1)$

RL for POMDPs

Approach 1: K-Markov

Approach 2: Recurrent N.N. (α^*)

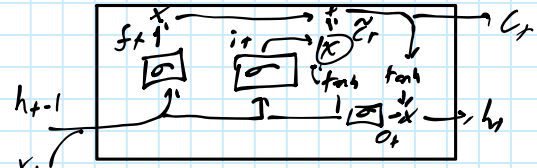


input: U_t State/output: x_t Cost: E_t Problem: Exploding/Vanishing Gradients

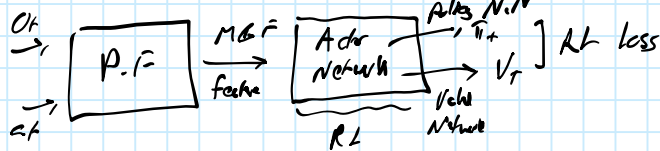
Recurrent N.N, Long-Short Term Memory (LSTM):

Input: x_t Cell state: C_t
Output: h_t

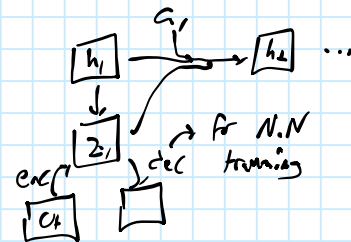
f_t : forget gate
 i_t : input gate
 o_t : output gate



Approach 3: PF & MGIF features



Approach 4:
Latent Representations (World Models)



MGIF feature: Order Invariant, recent governing fcn, Self transformer

Simple Games: Mathematical Formulas for rational interaction
Most Popular Solution Concept, Nash Equilibrium

Types of classmate's

Normal Form Games Bob

Allocators $\rightarrow$ MDP

Epistemic (Static) $\rightarrow$ RL

Epistemic (Dynamic) $\rightarrow$ POMDP

Interaction $\rightarrow$ Game

		S	W
Alice	S	4, 4	2, 3
	W	3, 2	1, 1

Left reward is left of me, right reward is for me.

Dominant Strategies

Definition:

- Action $a^i \in A^i$
- Joint Action $a = (a^1, \dots, a^k)$
- All other actions $a^{-i} = (a^1, \dots, a^{i-1}, a^{i+1}, \dots, a^k)$
- Reward $R^i(a)$ Joint Reward $R(a) = (R^1(a), \dots, R^k(a))$

Determine Best Response:

Action a^i is a determined best response to a^{-i} if

$$R^i(a^i, a^{-i}) \geq R^i(a^{i'}, a^{-i}) \quad \forall a^{i'}$$

Dominant (Pure) Strategy: Action a is a dominant strategy if it is a best response to every action taken by the other players.

Dominant Strategy Equilibrium: Every player plays a dominant strategy.

Prisoners Dilemma

P2

		S	T
P1	S	-1, -1 $\rightarrow$ -4, 0	
	T	0, -4 $\rightarrow$ -3, -3	

DSSE, DSSE is for both to testify, but a bad result for the criminals.

	CP	Don
CP	-5, -5	-1, 0
Don	0, -1	-4, -4

Two Pure Nash equilibrium
No DSSE

Pure Nash Equilibrium: All players play a determined best response. Which equilibrium is better?

Missile Defense Game

Defender

		CP	Don
Attacker	CP	-1, 1 $\leftarrow$ 1, -1	
	Don	1, -1 $\rightarrow$ -1, 1	

Drawing the arrows:

If defender goes CP, $-1 < 1$, attacker don

No Pure Nash $\rightarrow$ Equilibrium!

If Attacker goes CP, $-1 < 1$, defender CP

Mixed Nash Equilibrium

Notation for Mixed Strategies:

State: Attacker Joint
Action: $a^i \in A^i$ $a \in A$
Policy: $\pi^i(a^i)$ $\pi(a) = \prod \pi^i(a^i)$
Reward: $R^i(a^i)$ $R(a)$
Utility: $U^i(\pi) = \sum R^i(a^i) \pi(a^i)$ $U(\pi) = \sum R(a) \pi(a)$

The Player Zero Sum:

$$R^i(a^i) + R^k(a^k) = 0 \quad \forall a$$

Competition:

$$R^i(a^i) = R^k(a^k) \quad \forall i, j, a$$

Best Response: Given a joint policy of all other agents π^{-i} , a best response is a policy π^i that satisfies $U^i(\pi^i, \pi^{-i}) \geq U^i(\pi^{i'}, \pi^{-i}) \quad \forall \pi^{i'}$

A Nash Equilibrium is a joint policy in which all agents play a best response.

Rock-Paper-Scissors

P2

N.E: each player plays each action w/ $1/3$ probability

$$u^2(\pi) = u^2(\pi^1, \pi^2) = \sum_{a^1, a^2} \pi^1(a^1) \pi^2(a^2) R^2(a^1, a^2)$$

$$= \pi^1(R) \pi^2(R) (0) + \pi^1(R) \pi^2(P) (-1) + \dots$$

P1

	R	P	S
R	0,0	-1,1	1,-1
P	1,-1	0,0	-1,1
S	-1,1	1,-1	0,0

$$u^2(\pi^1, \pi^2) = \pi^2(P) (\pi^1(R) - \pi^1(S)) + \pi^2(R) (-\pi^1(P) + \pi^1(S)) + \pi^2(S) (\pi^1(P) - \pi^1(R))$$

Any policy is a B.R. to a $(1/3, 1/3, 1/3)$ strategy.

A deterministic policy is a B.R. to any strategy other than $(1/3, 1/3, 1/3)$

All finite games have at least one Nash equilibrium (might be mixed)

Calculating mixed Nash:

In mixed Nash Equilibrium, players must be indifferent between two or more actions

Algorithm that uses best response:

Iterated Best Response: Randomly cycle between agents who play the best response for the current policy (converges to Nash for certain narrow class of games)

P1

	Stay	Move
Stay	4,4	1,3
Move	3,1	2,2

P2 indifferent!

$$P2: \text{Stay } 4\pi^2(Stay) + 1\pi^2(Move) = V$$

$$P2: \text{Move } 3\pi^2(Stay) + 2\pi^2(Move) = V$$

$$\pi^2(Stay) = \pi^2(Move) = 1/2$$

P1 indifferent!

$\rightarrow$ P1's reward in Stay

$$P1: \text{Stay } 4\pi^1(Stay) + 1\pi^1(Move) = U$$

$$P1: \text{Move } 3\pi^1(Stay) + 2\pi^1(Move) = U \rightarrow \text{Solve}$$

$$\text{Probabilities: } \pi^1(Stay) + \pi^1(Move) = 1$$

$\rightarrow$ P1's reward in Move

Fictitious Play:

1) Estimate M.L. policies for all players $\pi^i(a^i) \propto N(x^i, a^i)$

2) Play best response to estimated policy.

P2

Back or Strengthening!

P1

	B	S
B	2,1	0,0
S	0,0	1,2

Correlated Equilibrium:

- A correlated joint policy is a single distribution over the joint actions of all agents.
- A correlated equilibrium is a correlated joint policy where no agent can increase their expected U. by deviating from their current policy.
- Easier to find than Nash equilibrium.

Overview:

- If every player is playing a best response, that joint policy is a Nash Equilibrium

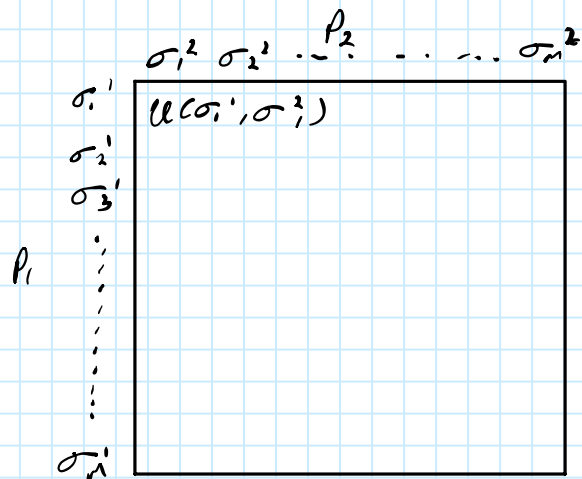
- Every finite game has at least one Nash equilibrium (Pure or mixed)

- Mixed Nash occurs when players are indifferent between two outcomes.

Dynamic Games: Markov Game Definition (I, S, A, T, R, γ)

- I : set of players
- S : state space, combined for all players
- $A = \{A^i\}_{i \in I}$ = joint action space
- T = transition distribution $T(S'|S, a)$ joint action
- R = joint reward fcn

Reduction to Mark Games:



$\sigma \rightarrow$ Deterministic Strategy

σ_i^1 , Player 1's deterministic policy

$$M = |A|^{|S|}$$

$\rightarrow$ Huge Matrix Game now

Best Response in a Markov Game

Given $MG(I, S, A, R, T, \gamma)$, π^i ,
 Calculate B.R. argmax $\sigma^i: U(\sigma^i, \pi^{-i})$

1) Formulate MDP

$$M^i = (S, A^i, T^i, R^i, \gamma)$$

$$T^i(S'|S, a^i) = \sum_{a^{-i} \in A^{-i}} T(S'|S, a^i, a^{-i}) \pi^{-i}(a^{-i}|S)$$

$$R^i(S, a^i) = \sum_{a^{-i} \in A^{-i}} R(S, a^i, a^{-i}) \pi^{-i}(a^{-i}|S)$$

2) Solve M^i (Value Iteration)

$\rightarrow$ Might not give Nash equilibrium

$$U^i(\sigma^1, \sigma^2) = \sum \gamma^t R^i(S_t, \sigma^1(S_t), \sigma^2(S_t))$$

Fictitious Play in a Markov Game

Loop:

1) Simulate w/ π^{BR}

2) Compute $NC_j(a^j, s)$ (N states reflect "all" past states)

3) $\pi^j(a^j, s) \propto NC_j(a^j, s) \forall j$

4) $\pi^{BR} \leftarrow \pi^j$ to π

$\rightarrow$ π converges to a Nash equilibrium in many cases, notably 2-player zero sum games